Multilayer local score with an entropy-barrier window

Does a second, larger-window LS layer raise Inception Score?

Tilt the window schedule, do not add a layer: CIFAR-10 FID 94 -> 83 and CelebA 79 -> 70 at the same memorization

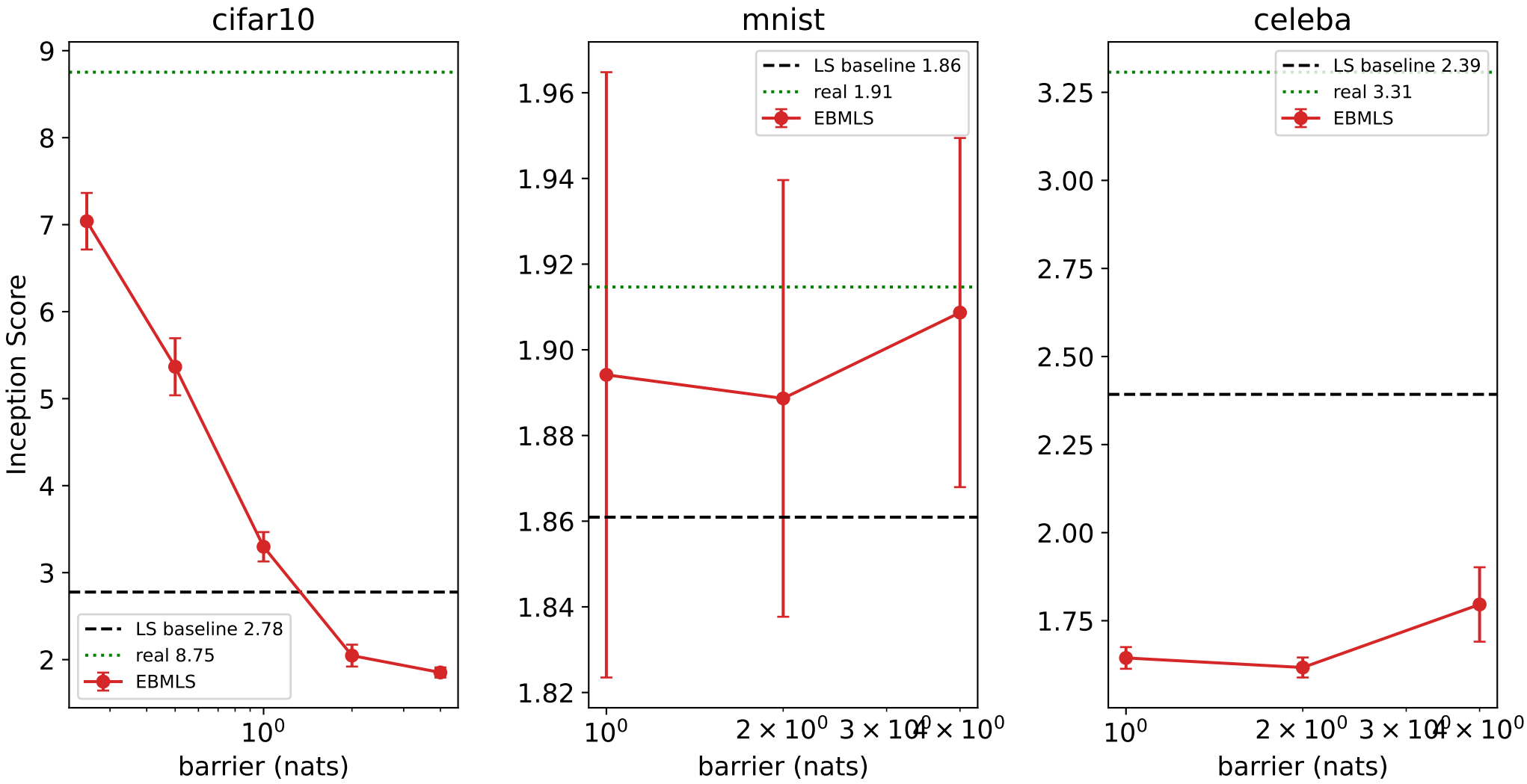
Layer 2 sees only layer 1's estimate, forms a posterior over all training images in a larger window, and adds the expected error of that estimate. The window at each step is the largest whose posterior entropy is still above a barrier — i.e. has not collapsed onto one image.

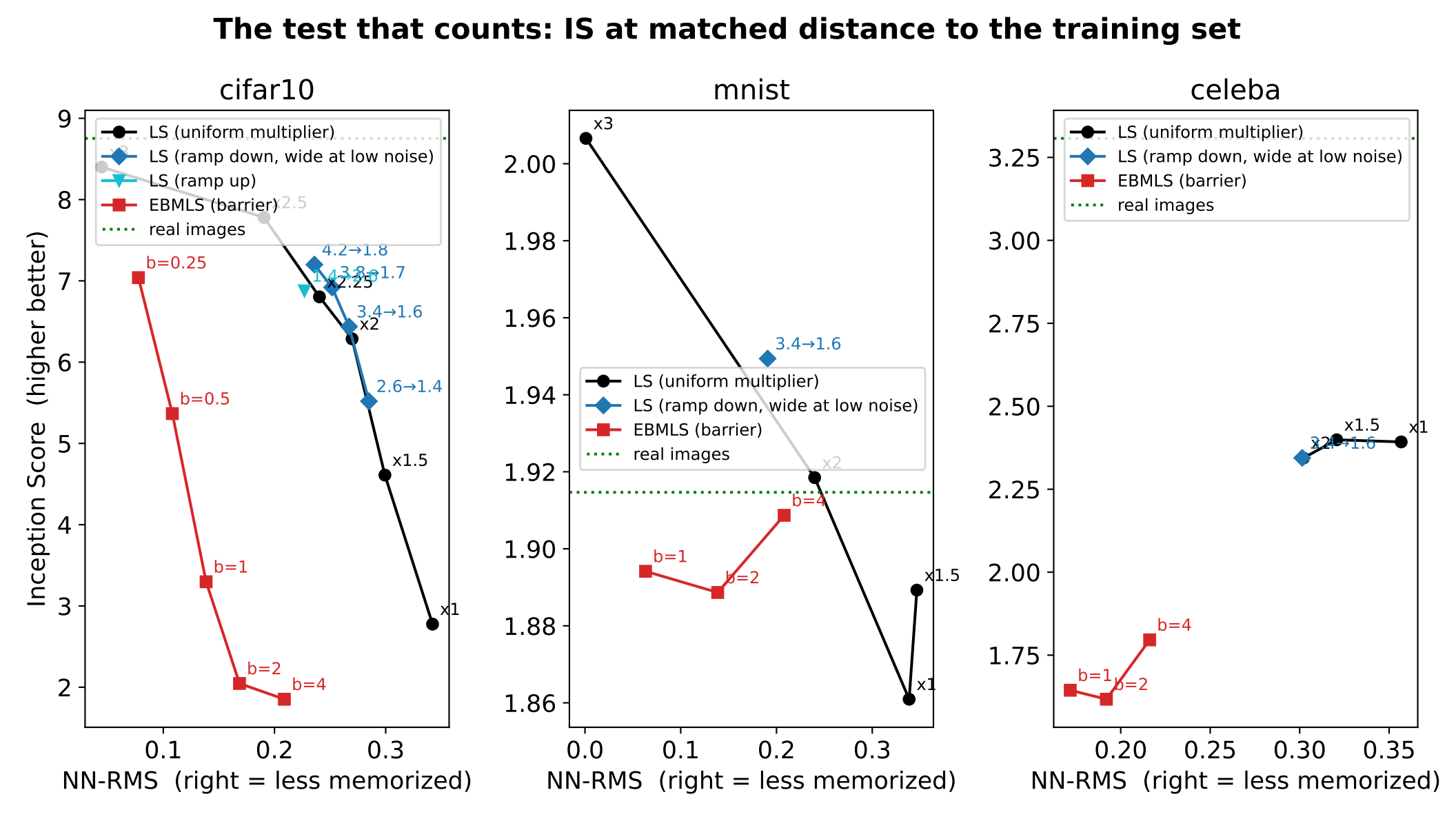
2026-10-21 · commit 5d6868f · seed 0 · 500 samples/config · IS over 5 splits · FID vs the full training set

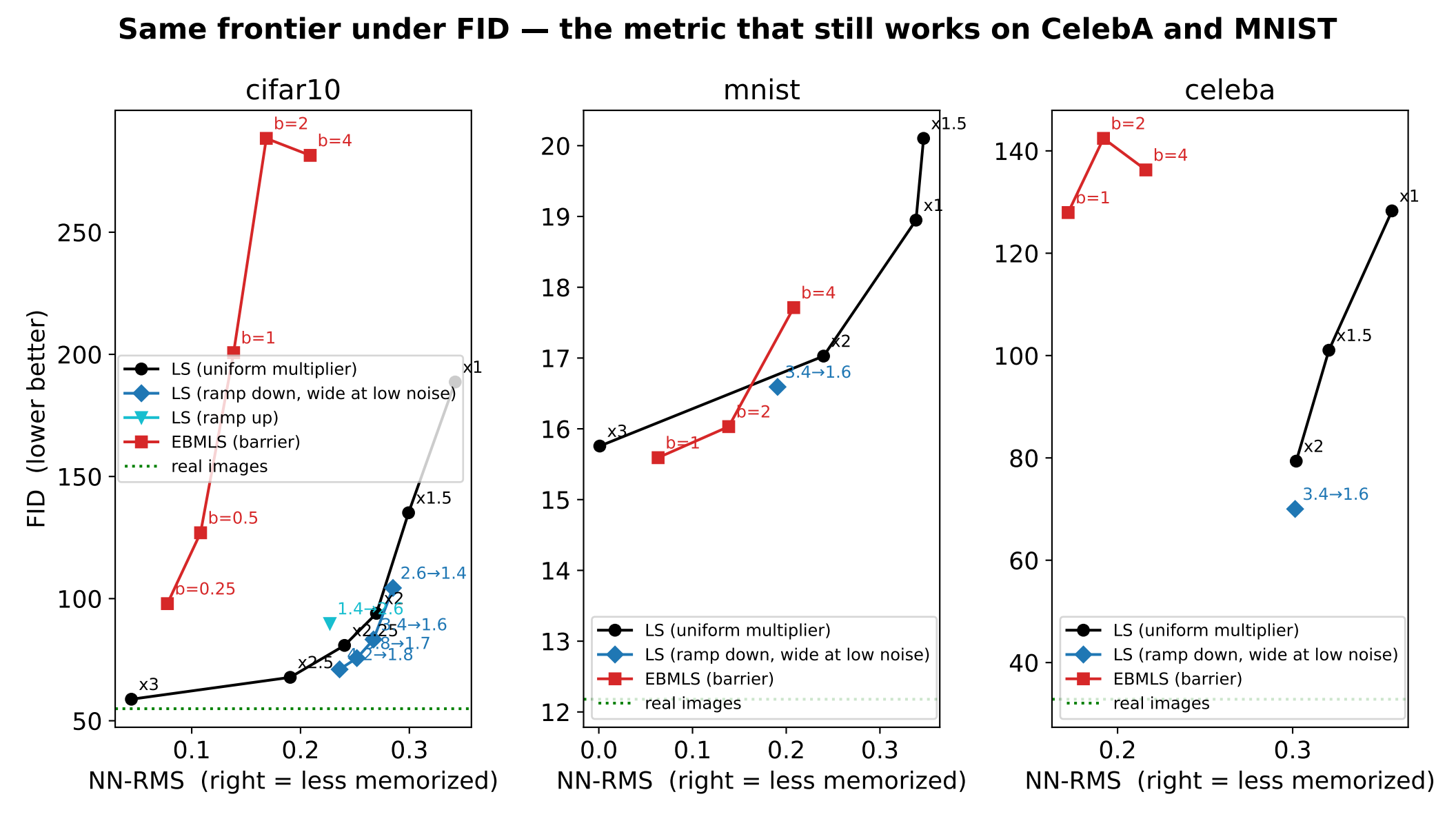
Metrics

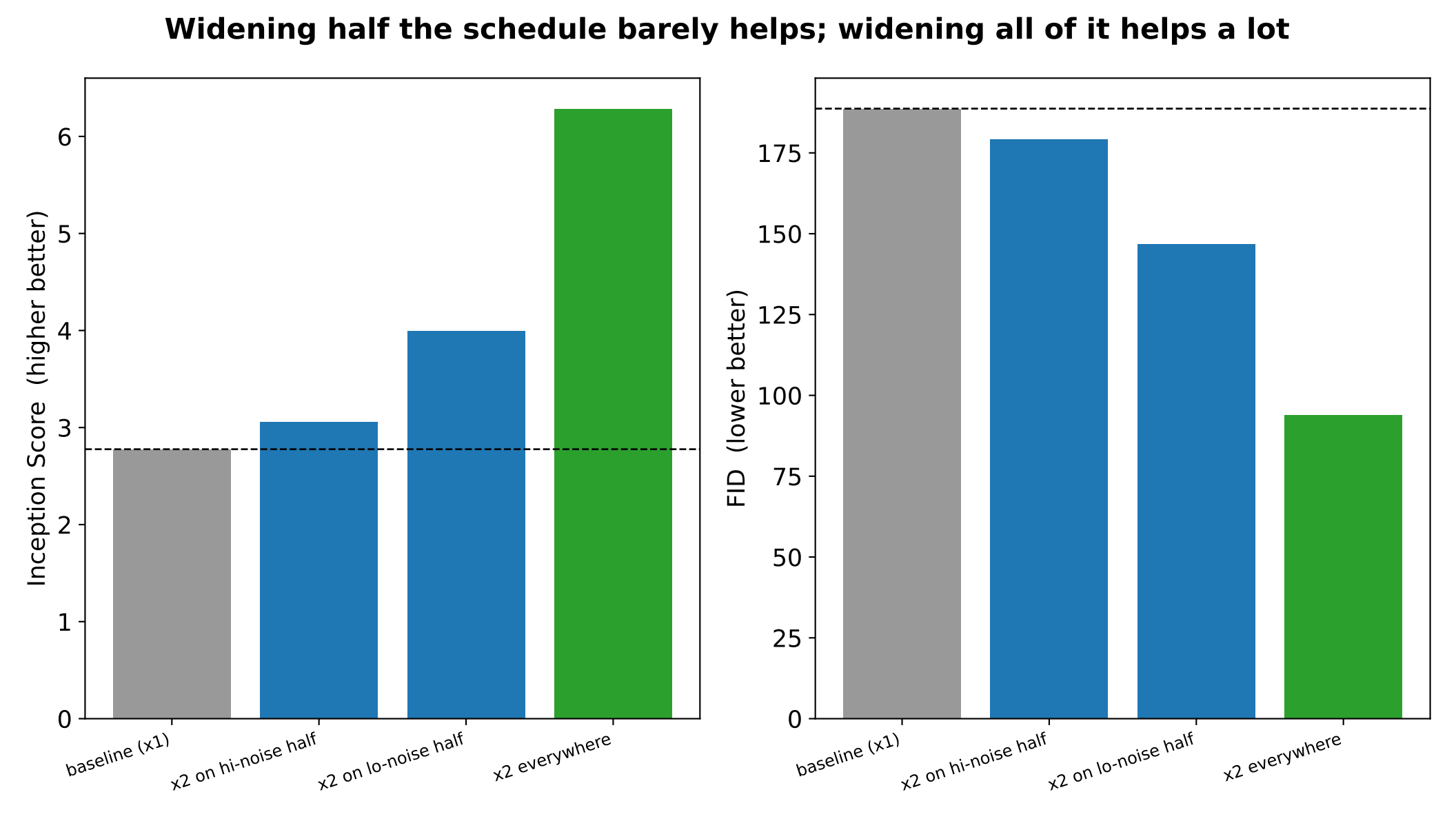
IS higher is better, FID lower is better; NN-RMS higher = less memorized

dataset	config	seed	IS	FID	NN-RMS	copy frac
celeba	real images	0	3.307 ± 0.119	32.8	—	—
celeba	LS	0	2.393 ± 0.043	128.3	0.357	0.000
celeba	LS (windows x1.5)	0	2.399 ± 0.214	101.0	0.321	0.000
celeba	LS (windows x2)	0	2.344 ± 0.090	79.4	0.302	0.000
celeba	LS (ramp x3.4→x1.6)	0	2.344 ± 0.193	70.0	0.301	0.000
celeba	EBMLS barrier 1	0	1.644 ± 0.031	127.9	0.172	0.000
celeba	EBMLS barrier 2	0	1.617 ± 0.028	142.4	0.192	0.000
celeba	EBMLS barrier 4	0	1.796 ± 0.106	136.3	0.216	0.000
cifar10	real images	0	8.753 ± 0.465	54.9	—	—
cifar10	LS	0	2.777 ± 0.157	188.7	0.342	0.000
cifar10	LS	1	2.857 ± 0.061	190.4	0.343	0.000
cifar10	LS (windows x1.5)	0	4.610 ± 0.215	135.1	0.299	0.000
cifar10	LS (windows x2)	0	6.286 ± 0.563	94.0	0.270	0.000
cifar10	LS (windows x2)	1	6.073 ± 0.590	93.0	0.274	0.000
cifar10	LS (windows x2, hi-noise half)	0	3.061 ± 0.179	179.2	0.286	0.000
cifar10	LS (windows x2, lo-noise half)	0	3.997 ± 0.252	146.8	0.299	0.000
cifar10	LS (windows x2.25)	0	6.803 ± 0.365	80.8	0.240	0.028
cifar10	LS (windows x2.5)	0	7.780 ± 0.360	67.8	0.191	0.146
cifar10	LS (windows x3)	0	8.401 ± 0.467	58.8	0.045	0.514
cifar10	LS (ramp x2.6→x1.4)	0	5.519 ± 0.456	104.4	0.285	0.000
cifar10	LS (ramp x3.4→x1.6)	0	6.437 ± 0.310	83.3	0.267	0.002
cifar10	LS (ramp x3.4→x1.6)	1	6.314 ± 0.284	86.1	0.271	0.000
cifar10	LS (ramp x3.8→x1.7)	0	6.919 ± 0.417	75.6	0.252	0.034
cifar10	LS (ramp x4.2→x1.8)	0	7.200 ± 0.573	71.0	0.236	0.092
cifar10	LS (ramp x1.4→x2.6)	0	6.877 ± 0.297	89.8	0.227	0.016
cifar10	EBMLS barrier 0.25	0	7.040 ± 0.325	97.9	0.078	0.196
cifar10	EBMLS barrier 0.5	0	5.367 ± 0.328	127.0	0.108	0.058
cifar10	EBMLS barrier 1	0	3.298 ± 0.169	200.7	0.139	0.004
cifar10	EBMLS barrier 2	0	2.046 ± 0.126	288.4	0.169	0.000
cifar10	EBMLS barrier 4	0	1.853 ± 0.058	281.5	0.209	0.000
mnist	real images	0	1.915 ± 0.027	12.2	—	—
mnist	LS	0	1.861 ± 0.062	18.9	0.339	0.000
mnist	LS (windows x1.5)	0	1.889 ± 0.054	20.1	0.347	0.000
mnist	LS (windows x2)	0	1.918 ± 0.091	17.0	0.240	0.078
mnist	LS (windows x3)	0	2.007 ± 0.082	15.8	0.001	0.846
mnist	LS (ramp x3.4→x1.6)	0	1.949 ± 0.082	16.6	0.191	0.250
mnist	EBMLS barrier 1	0	1.894 ± 0.071	15.6	0.063	0.460
mnist	EBMLS barrier 2	0	1.889 ± 0.051	16.0	0.139	0.160
mnist	EBMLS barrier 4	0	1.909 ± 0.041	17.7	0.208	0.008

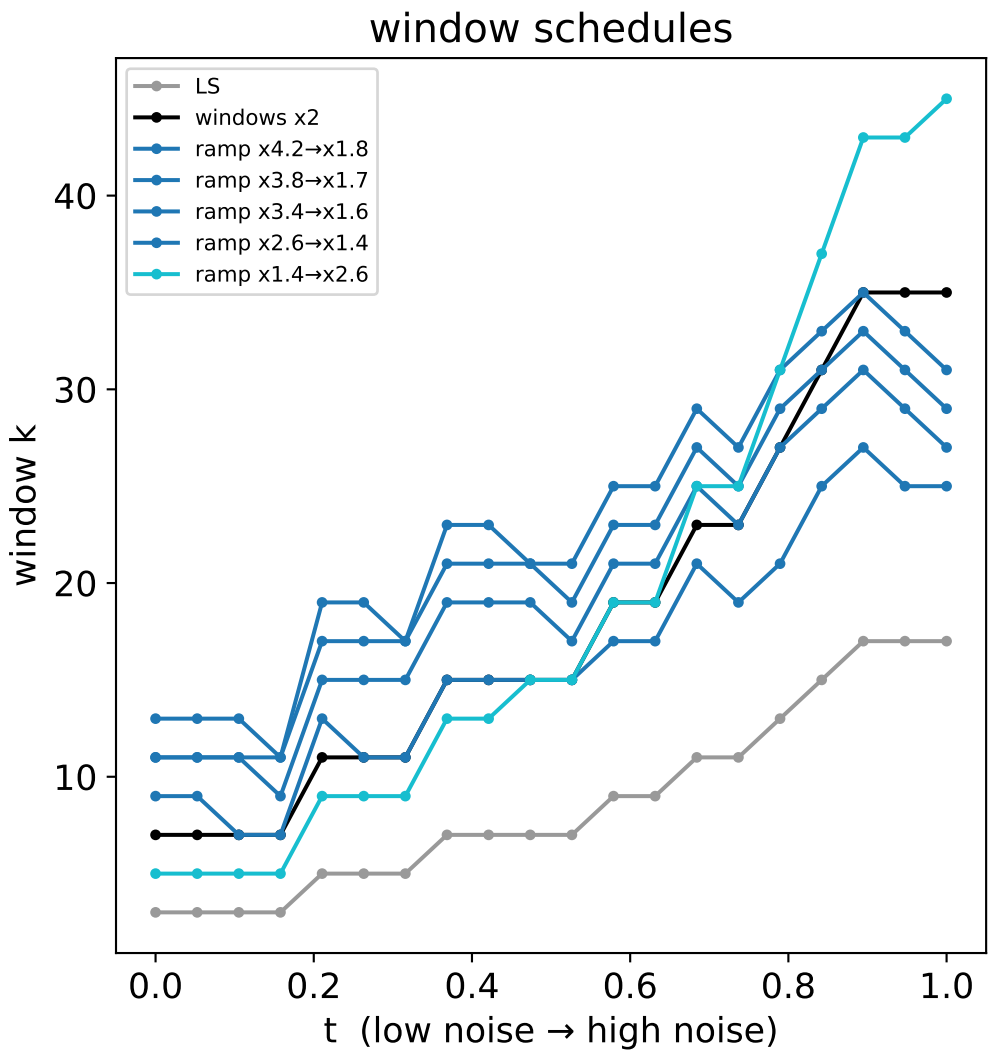

Inception Score vs entropy barrier
cifar10
mnist
celeba
Inception Score
barrier (nats)
LS baseline 2.78
real 8.75
EBMLS
LS baseline 1.86
real 1.91
EBMLS
LS baseline 2.39
real 3.31
EBMLS

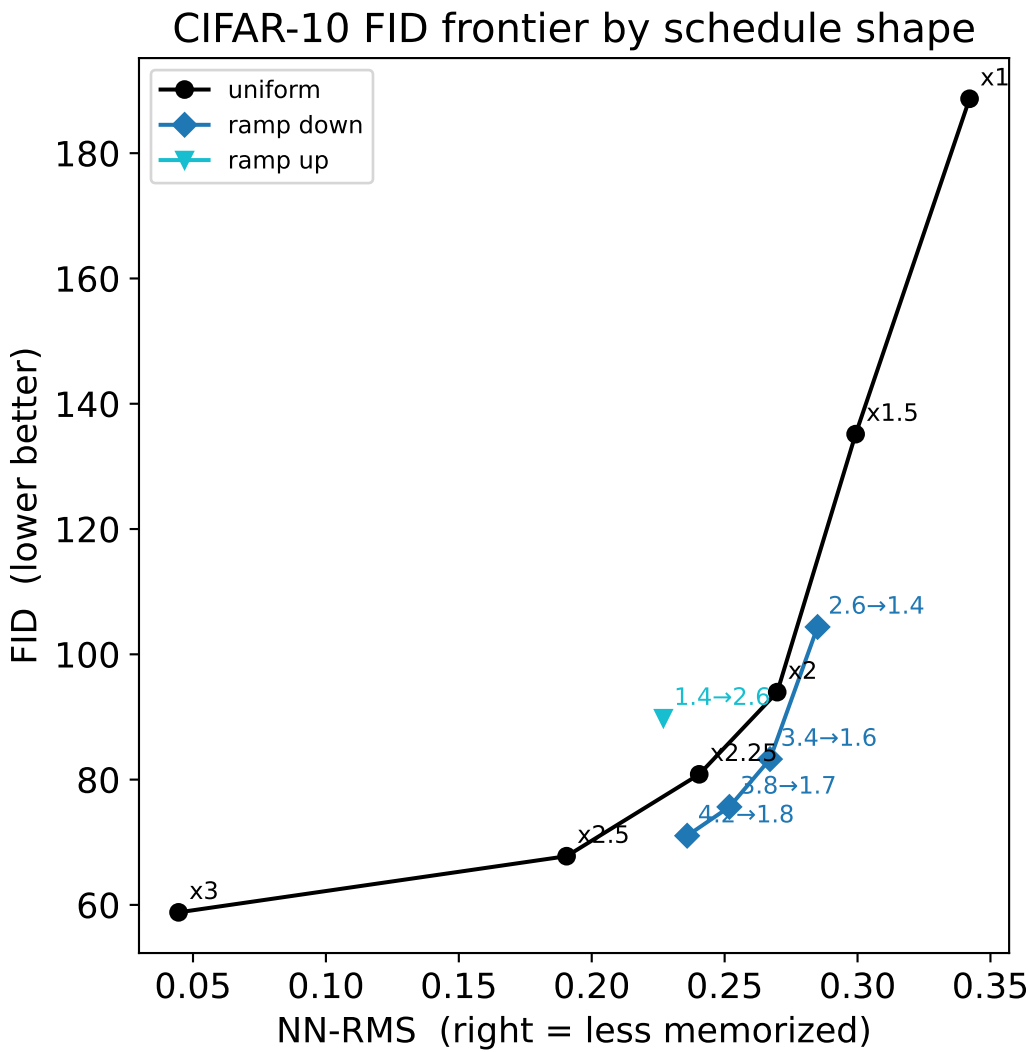

The test that counts: IS at matched distance to the training set
cifar10
mnist
celeba
Inception Score (higher better)
NN-RMS (right = less memorized)
LS (uniform multiplier)
LS (ramp down, wide at low noise)
LS (ramp up)
EBMLS (barrier)
real images
b=0.25
b=0.5
b=1
b=2
b=4
x3
x2.5
x2.25
x2
x1.5
x1
4.2→1.8
3.8→1.7
3.4→1.6
2.6→1.4

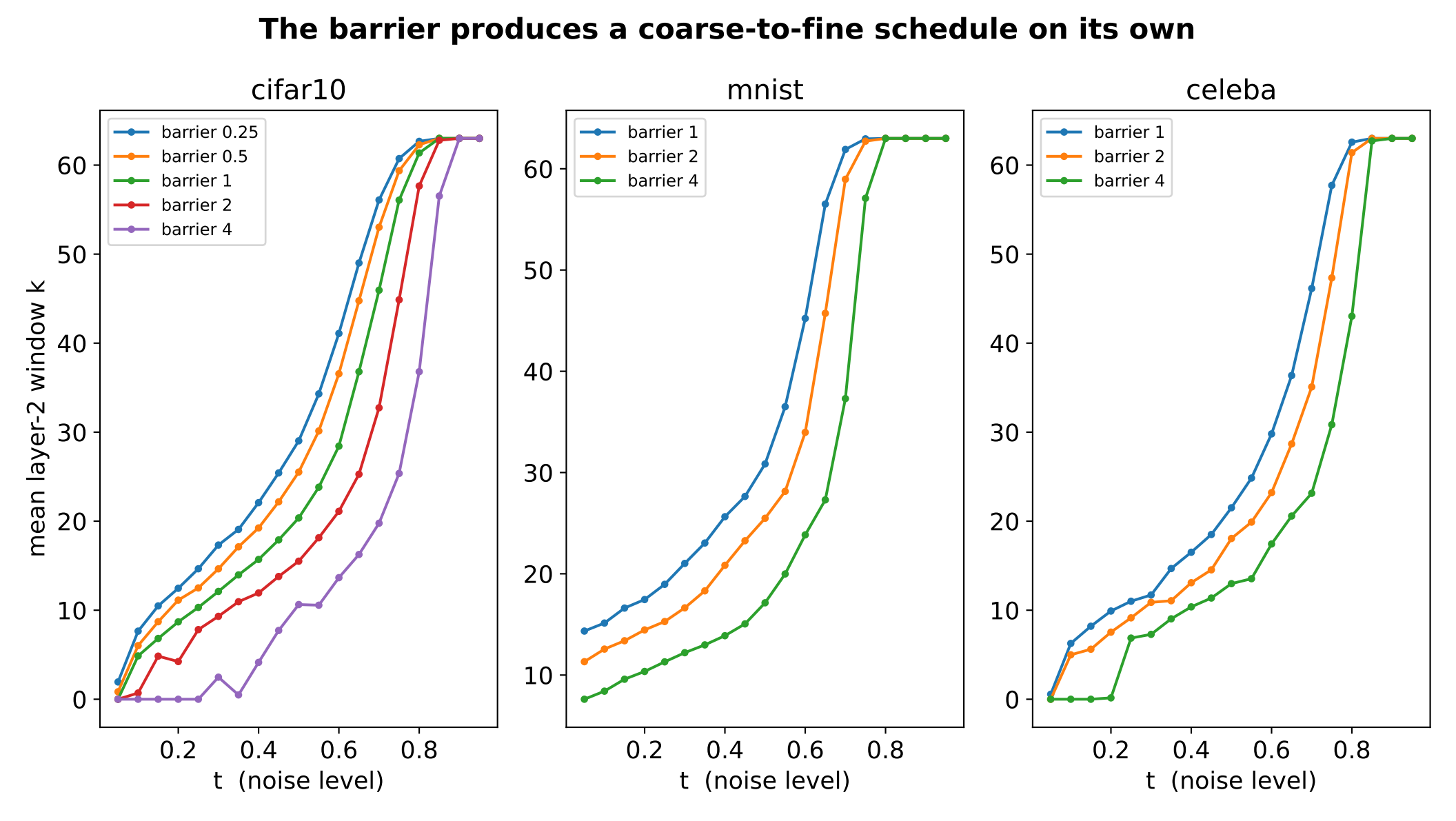

Same frontier under FID — the metric that still works on CelebA and MNIST
cifar10
mnist
celeba
FID (lower better)
NN-RMS (right = less memorized)
LS (uniform multiplier)
LS (ramp down, wide at low noise)
LS (ramp up)
EBMLS (barrier)
real images
b=2
b=4
b=1
b=0.5
b=0.25
x1
x1.5
x2
x2.5
x3
2.6→1.4
3.4→1.6
1.4→2.6
2.8→1.7
2→1.8


Widening half the schedule barely helps; widening all of it helps a lot
Inception Score (higher better)
0
1
2
3
4
5
6
baseline (x1)
x2 on hi-noise half
x2 on lo-noise half
x2 everywhere
FID (lower better)
0
25
50
75
100
125
150
175
baseline (x1)
x2 on hi-noise half
x2 on lo-noise half
x2 everywhere

Tilting the schedule beats widening it uniformly

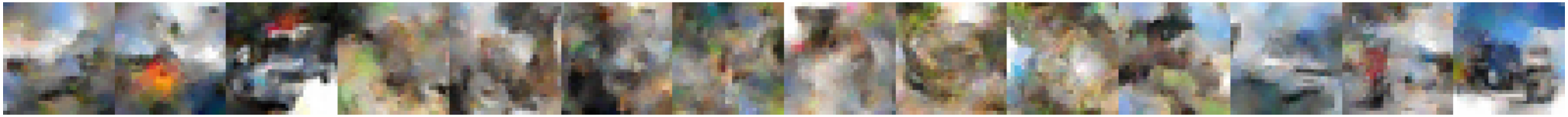





The barrier produces a coarse-to-fine schedule on its own
cifar10
mnist
celeba
barrier 0.25
barrier 0.5
barrier 1
barrier 2
barrier 4
mean layer-2 window k
t (noise level)
0
10
20
30
40
50
60
0.2
0.4
0.6
0.8

cifar10 samples: baseline, best non-copying, best EBMLS, and the memorizing end

LS — IS 2.78, NN-RMS 0.342, copies 0.0%



LS (ramp x3.4→x1.6) — IS 6.44, NN-RMS 0.267, copies 0.2%



EBMLS barrier 0.25 — IS 7.04, NN-RMS 0.078, copies 19.6%



LS (windows x3) — IS 8.40, NN-RMS 0.045, copies 51.4%



Where this leaves us

1. The entropy-barrier multilayer layer does not pay off. It is dominated by plain LS with wider windows at every matched memorization level, on both metrics. FID sharpens this: barrier 1 beats the LS baseline on IS (3.30 vs 2.78) but is clearly worse on FID (201 vs 189), so its apparent IS gain was not a distribution gain.

   Why: layer 2's per-pixel weights sum to 1, so x1 + d2 = sum_n w_n(p) x0_n(p) exactly. Layer 2 replaces layer 1's estimate rather than refining it, and widening its window walks straight into copying. (The layer-2 posterior is under revision -- hshunt is re-deriving the correct posterior for a training-set image -- which will change these numbers but not this point.)

2. The result came from the control: the per-step window schedule, not the layer count, is what moves the metrics. Uniformly scaling the windows x2 improves every dataset with zero training-set copies:

   CIFAR-10 IS 2.78 -> 6.29, FID 189 -> 94
   CelebA IS 2.39 -> 2.34, FID 128 -> 79
   MNIST IS 1.86 -> 1.92, FID 19.0 -> 17.0